SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	2024	**Date:**	1/8/2026

Code of Conduct

I ____karthik/192472281______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___karthik______________________

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, RuleBased Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4

English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

2.Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

3.Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

4.Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

5.Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

1. Morphological Analysis Pipeline

Input Words: connected, connecting, connection

Aim

Design a morphological analysis pipeline for document indexing by decomposing words into root and suffix, identifying inflectional/derivational forms, and normalizing them.

Word	Root	Suffix	Type	Morphological Breakdown	Normalized Output
connected	connect	-ed	Inflectional (Past Tense)	connect + ed	**connect**
connecting	connect	-ing	Inflectional (Present Participle)	connect + ing	**connect**
connection	connect	-ion	Derivational (Noun Formation)	connect + ion	**connect**

Morphological Analysis Pipeline

1. Input word.
2. Detect suffix using rule-based analyzer.
3. Separate root and suffix.
4. Identify suffix type (Inflectional/Derivational).
5. Normalize all words to the root **connect**.
6. Store normalized form for indexing.

2. Morphological Parsing Module

Input Words: unhappy, happiness, happily

Aim

Design a rule-based morphological parser for sentiment analysis.

Word	Prefix	Root	Suffix	Type	Morphological Breakdown
unhappy	un-	happy	—	Derivational	un + happy

Word	Prefix	Root	Suffix	Type	Morphological Breakdown
happiness	—	happy	-ness	Derivational	happy + ness
happily	—	happy	-ly	Derivational	happy + ly

Semantic Consistency

- Base word: **happy**
- unhappy → negative sentiment
- happiness → noun form
- happily → adverb form

Normalized Root

Happy

3. Stemming-Based Preprocessing Module

Input Words: played, player, playing

Aim

Apply stemming rules to identify the common base word.

Word	Affix Removed	Stem	Classification
played	-ed	play	Inflectional
player	-er	play	Derivational
playing	-ing	play	Inflectional

Morphological Rules

- Remove **-ed**
- Remove **-ing**
- Remove **-er**

Final Stem

Play

4. Finite-State Morphological Parsing

Input Words: writes, writing, written

Common Root

write

Word	**State Transition**	**Morphological Breakdown**	**Pattern**	**Normalized Output**
writes	Write → Add **-s**	write + s	Regular Inflection	**write**
writing	Write → Drop **e** → Add **-ing**	write → writing	Regular Inflection	**write**
written	Write → Irregular change → written	write → written	Irregular Inflection	**write**

Finite-State Transitions

- Start → **write**
- write + **s** → writes
- write − **e** + **ing** → writing
- write → written (irregular transition)

Final Normalized Root

Write

5. Porter Stemmer-Based Preprocessing

Input Words: relational, relation, relate

Aim

Apply Porter Stemmer rules to obtain the common stem.

Word	Stemming Steps	Intermediate Form	Final Stem
relational	relational → remove **-ational** → relate → remove **-e**	relate	**relat**
relation	relation → remove **-ion**	relat	**relat**
relate	relate → remove **-e**	relat	**relat**

Porter Stemmer Process

1. Detect derivational suffix.
2. Remove suffix according to Porter rules.
3. Produce the common stem.
4. Store the stem for document retrieval.

Final Stem

Relat

Faculty In-charge	Course Coordinator	Program Director

Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

Rubrics for Calculation

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	**Problem Understanding**	**Method Selection**	**Solution Process**	**Accuracy of Calculation**	**Interpretation of Results**	**Sub. Total**	**Total %**
1							
2							
3							
4							
5							